

DESIGN EVALUATION 2

XR Study Whiteboard — Iteration 2

Name	Date	Prototype / round	Participants / device
Andhika Nayaka Arya Wibowo	3 October 2026	XR Study Whiteboard / 2	5 students / VR headset + 2 controllers

Objective and validation metrics

The evaluation tested whether the revised prototype supports a complete headset-based classroom workflow with minimal assistance: navigating between a shared whiteboard and student desks, viewing the teaching board while seated, taking personal notes, controlling both drawing surfaces, and recording and replaying whiteboard actions. Success thresholds were defined in the Iteration 2 plan before testing, so the evaluation checks the evidence against those targets rather than redefining success after the sessions.

Testing aim	Predeclared success measure
Shared seated view	At least 4/5 see the relevant board content and full paper from the seated pose.
Paper interaction	At least 4/5 discover Pencil; required strokes are reliable, correctly oriented, and do not continue after release.
Lesson-to-notes workflow	At least 4/5 copy the date, a line, and a circle while seated and rate comfort 4–5/5.
Navigation and tracking	Both controllers remain aligned; at least 4/5 rate movement predictability 4–5/5.
Record and replay	At least 4/5 complete Record–draw–Stop–Play and understand that actions, not audio, are replayed.
Overall confidence	Median $\geq 4/5$ with no severe comfort, tracking, data-loss, or blocking issue.

Method and evidence

Five new university-student participants completed the same moderated workflow in class using the Unity VR build, a headset, and two physical controllers. The sessions used think-aloud prompts, task observation, six 1–5 ratings, and a final reflection. All participants used physical controllers, so the desktop-simulator T/Y task was correctly marked not applicable.

P01 (Kai) has a voice recording and transcript; P02 (Syifa) has a camera recording and transcript. P03 (Amy), P04 (Serafina), and P05 (Safira) were not audio/video recorded and are represented only by in-class observation notes and ratings. No quotations are attributed to P03–P05. This separation avoids presenting notes as recorded testimony.

Factual results

Observed task	P01	P02	P03	P04	P05	Outcome
Shared seated view	Pass	Pass	Pass	Pass	Pass	5/5
Copy board content to paper	Pass	Pass	Pass	Pass	Pass	5/5
Paper write / erase / clear	Pass	Pass	Pass	Pass	Pass	5/5
Move and return	Pass	Pass	Pass	Pass	Pass	5/5
Whiteboard workflow	Pass	Pass	Pass	Pass	Pass	5/5
Record–Stop–Play	Pass	Pass	Pass	Pass	Pass	5/5
Physical controller tracking	Pass	Pass	Pass	Pass	Pass	5/5

Results, synthesis, and insights

Post-test measure	P01	P02	P03	P04	P05	Median
Seated view comfort	5	5	5	5	5	5/5
Paper drawing reliability	5	5	5	5	5	5/5
Paper control clarity	5	5	5	5	5	5/5
Movement predictability	5	5	5	5	5	5/5
Recording control clarity	5	5	5	5	5	5/5
Overall confidence	5	5	5	5	5	5/5

All five sessions were completed. Every participant passed the observed headset workflow and all six rating medians were 5/5. No nausea, dizziness, eye strain, tracking loss, data loss, or technical blocker was recorded. P01 said the seated task was “pretty easy” and could look between board and paper; P02 identified the pop-up as a control guide and used it again when needed. P01 and P02 both reported that the physical controllers remained responsive and aligned.

Analysis and insights

Pattern grounded in evidence	Interpretation for the concept
The shared-view task passed 5/5; participants copied board content while remaining at a desk; comfort median was 5/5.	The revised seat pose supports the concept’s central learning loop: attend to shared instruction, then externalise notes on a private surface without leaving the study position.
Paper write, erase, and clear passed 5/5; reliability and clarity medians were 5/5.	Aligning paper coordinates, fixing the paper in place, and using the right controller as the writing tool removed the Round 1 ambiguity between manipulating an object and writing on it.
Movement and controller tracking passed 5/5; movement predictability median was 5/5.	Destination shortcuts and coupled headset/controller movement created a stable mental model. Navigation now supports the task rather than interrupting it.
Record–Stop–Play passed 5/5; P01 predicted action replay and P02 confirmed the replay matched expectations.	Action-based recording is understandable when its status and sequence are visible. The concept does not need microphone recording to communicate this feature.
P01 needed brief familiarisation; P02 understood the guide; P04 notes recommend a short introduction.	The remaining issue is onboarding, not core interaction failure. The guide should stay concise, dismissible, and available on demand.

Evaluation of aims

Aim / target	Evidence and judgement
Shared seated view $\geq 4/5$	Validated: 5/5 passed; all five comfort ratings were 5/5.
Paper tool discovery $\geq 4/5$ within 15 s	Functionally supported: 5/5 used the controls. Exact 15-second unassisted timing was not recorded, so that narrow threshold remains unverified.
Paper drawing $\geq 90\%$ acceptable strokes	Functionally supported: 5/5 completed write/erase/clear and rated reliability 5/5. Raw stroke/defect counts were not captured, so 90% cannot be recalculated.
Lesson-to-notes $\geq 4/5$	Validated: 5/5 copied the required content and reported comfortable shared viewing.
Navigation/tracking and predictability	Validated: 5/5 passed; median predictability 5/5; no controller separation observed.
Recording $\geq 4/5$	Validated: 5/5 completed the sequence; recorded evidence confirms the intended action-replay model.
Overall confidence $\geq 4/5$; no severe issue	Validated: median 5/5; 0 severe comfort issues, blockers, incomplete sessions, or tracking failures.
Simulator T/Y selection	Not evaluated: all sessions used physical VR controllers. This aim cannot be claimed from participant evidence.

Concept iteration and design decision

Insight	Iteration / affirmation	Reasoning
The seated two-surface workflow passed 5/5.	Keep the classroom, shared whiteboard, fixed personal paper, and board-visible seat anchors.	These elements directly support the intended follow-and-take-notes behaviour.
Brief first-use familiarisation remained.	Retain the modern controller guide; show it at launch, allow A to close/reopen it, and keep X = whiteboard and Y = next table visible.	P02 used the guide successfully; P01 and P04 evidence shows that a short introduction still helps.
Paper interaction was reliable and clear.	Keep paper ungrabbable and reserve the right trigger for tool selection and writing.	This prevents the selection gesture from moving the writing surface and preserves a clear role for each controller.
No core failure repeated; optional features were suggested.	Prioritise saved notes and additional classroom environments only after an independent usability retest.	P03–P05 notes frame these as future enhancements, not fixes required for task completion.

Updated concept description

The XR Study Whiteboard is a headset-based remote-learning classroom in which a learner uses the left controller to navigate, the right controller to select tools and write, and a concise in-world guide for first-use support. A dedicated whiteboard shortcut places the learner at an operable standing height; table shortcuts place them where the teaching board and fixed note paper are visible together. The learner can write, erase, and clear personal notes, use the shared whiteboard, and record and replay timed whiteboard actions. Iteration 2 therefore preserves the classroom presence and two-surface study model while removing the visibility, paper, navigation, and controller problems found in Round 1.

Critical reflection and next steps

The standard task order and participant matrix made repeated outcomes easy to compare, and recording two sessions provided traceable examples of expectations and behaviour. The method also has limits. Three sessions rely on facilitator notes rather than recordings; the facilitator demonstrated or prompted some controls; all ratings reached the scale ceiling; exact discoverability times and stroke-defect counts were not captured; and the convenience sample was small. The findings therefore validate usability of this build for the tested workflow, but do not establish long-term learning effectiveness, independent first-use performance, or simulator behaviour.

- Next test: recruit at least five new participants and give only the in-headset guide, with no control demonstration before the discoverability measure.
- Capture time to first Pencil selection, prompt level, required-stroke counts, controller tracking loss, and a short standard comfort measure.
- Run a longer teacher-to-student note-taking scenario and test whether notes can be saved or exported without disrupting the current workflow.
- Test the simulator T/Y control separately and report it as technical/simulator evidence, not headset participant evidence.

Final evaluation

Decision	Justification
VALIDATED for the tested Iteration 2 workflow	Applicable core targets were met by 5/5 participants with no severe comfort, tracking, data-loss, or blocking issue. Keep the core interaction design and treat onboarding and optional persistence/environment features as bounded future refinements. Exact timed discovery, stroke-rate, simulator, and educational-outcome claims remain outside the evidence.

APPENDICES — RAW AND SUPPORTING EVIDENCE (OUTSIDE THE THREE-PAGE ASSESSED RESPONSE)

Appendix A — Evidence register and participant records

All sessions used the Unity XR Study Whiteboard Iteration 2 headset build and two physical controllers. Names are retained because they were supplied with the session records; the assessed response uses participant IDs.

ID	Participant	Evidence	VR experience	Outcome
P01	Kai	Voice recording + transcript	Not recorded	Completed
P02	Syifa	Camera recording + transcript	Occasional	Completed
P03	Amy	In-class notes only; no audio/video	Occasional	Completed
P04	Serafina	In-class notes only; no audio/video	None	Completed
P05	Safira	In-class notes only; no audio/video	Occasional	Completed

Recording evidence

P01 voice recording: [Google Drive evidence](#)

P02 camera recording: [Google Drive evidence](#)

P03–P05 were not audio/video recorded. Their records are observation summaries and ratings, not transcripts.

Participant-level task matrix

Task	P01	P02	P03	P04	P05
Explore	Pass	Pass	Pass	Pass	Pass
Seat and shared view	Pass	Pass	Pass	Pass	Pass
Follow and take notes	Pass	Pass	Pass	Pass	Pass
Paper tools and strokes	Pass	Pass	Pass	Pass	Pass
Move and return	Pass	Pass	Pass	Pass	Pass
Whiteboard workflow	Pass	Pass	Pass	Pass	Pass
Record and replay	Pass	Pass	Pass	Pass	Pass
Physical controller tracking	Pass	Pass	Pass	Pass	Pass
Simulator T/Y choice	N/A	N/A	N/A	N/A	N/A

Appendix B — Ratings and notes-only session summaries

Measure	P01	P02	P03	P04	P05	Median
Seated view comfort	5	5	5	5	5	5/5
Paper drawing reliability	5	5	5	5	5	5/5
Paper control clarity	5	5	5	5	5	5/5
Movement predictability	5	5	5	5	5	5/5
Recording control clarity	5	5	5	5	5	5/5
Overall confidence	5	5	5	5	5	5/5

P03 — Amy (notes only)

All planned headset tasks passed on the first recorded attempt. The notes report stable navigation, correct Pencil/Eraser/Clear Paper use, successful Record–Stop–Play, 5/5 on all six measures, 10/10 satisfaction, and interest in remote learning. Optional additional classroom features were suggested; no core usability change was requested.

P04 — Serafina (notes only)

All planned headset tasks passed on the first recorded attempt. The notes report comfortable shared viewing, stable controller tracking, successful drawing and replay, 5/5 on all six measures, 10/10 satisfaction, and interest in remote learning. A short introduction for first-time VR users was the only suggested refinement.

P05 — Safira (notes only)

All planned headset tasks passed on the first recorded attempt. The notes report comfortable shared viewing, reliable paper and whiteboard interaction, stable navigation, successful replay, 5/5 on all six measures, 10/10 satisfaction, and interest in remote learning. Saved notes or more classroom environments were suggested as optional future features.

Appendix C — Recorded interview transcripts

The following transcripts reproduce the two recorded sessions. A moderator note is retained where P01 verbally referred to a mouse despite the session being conducted with a headset and physical controllers.

P01 — Kai: voice-recorded transcript

Nayaka: Good afternoon. My name is Nayaka, and I am going to guide you through testing my XR Study Whiteboard. Before we start, may I know your name?

Kai: Yeah, my name is Kai.

Nayaka: Nice to meet you.

Kai: Nice to meet you too.

Nayaka: Before we start, you are using the Unity VR build with the headset and two physical controllers. Can you see both controllers in the environment?

Kai: Yeah, I can see both of them and they are following my hands.

Nayaka: First, can you tell me what you see in the XR environment?

Kai: It looks like I am in a classroom right now. I can see a whiteboard. It says XR Study Whiteboard and Study Session. I can also see black, red, marker, eraser, Ready to Record, and Play.

Nayaka: Can you see all the buttons?

Kai: Yeah. Whiteboard, Stand, Jump, Center Table.

Nayaka: Can you try pressing one of them?

Kai: Yeah.

Nayaka: What do you think these buttons do? What were you expecting?

Kai: Initially, I did not know what to do. But now that I have pressed it, I can see that it teleports me to one of the tables depending on which one I press. If I press Stand, it puts me back.

Nayaka: When you move, do both VR controllers stay in the correct position with your hands?

Kai: Yeah, both of them stay with me.

Nayaka: Okay. Can you move to the whiteboard?

Kai: Yeah.

Nayaka: After teleporting to the whiteboard, what do you think you can do here?

Kai: Looking at the marker and everything, I expect that I could act like a teacher and write something on the whiteboard.

Nayaka: Okay. Can you try it?

Kai: Hello.

Nayaka: What do you think the eraser does?

Kai: I imagine it erases whatever I just wrote.

Nayaka: Nice. What other controls can you see?

Kai: Clear Board.

Nayaka: What do you expect that to do?

Kai: Clear my whole board.

Nayaka: Yes, that is right. What about changing colours?

Kai: Yeah, I can change the colour.

Nayaka: Do you see any other buttons?

Kai: There is a Record button.

Nayaka: What do you think that does?

Kai: It says Ready to Record, and that it records board actions and timings. I am assuming it records what I write. Then when I press Play, it will repeat what I wrote.

Nayaka: Can you try it?

Kai: For example, if I write “Hello”, then press Stop and Play, I would imagine it clears the board first and then writes “Hello” again.

Nayaka: Yes, that is right.

Kai: Oh, okay.

Nayaka: Great. We have finished testing the whiteboard. Can you now try the features on the desk?

Kai: Yep. Now I am at one of the desks. I am pressing the left click button on the paper to write “Hello”. I am going to press Eraser, and I expect it to erase what I wrote.

Nayaka: Yes.

Kai: Now I am going to write a lot on the paper. Then I will press Clear Paper. I expect it to clear the paper.

Nayaka: Okay. While you are at the desk, can you see the paper and the teaching board comfortably from the seated position?

Kai: Yeah. I can look between the paper and the board without needing to move away from the desk.

Nayaka: Can you copy the date, horizontal line, and circle from the teaching board onto the paper?

Kai: Yeah.

Nayaka: How did that feel?

Kai: Pretty easy. I can look at the board and then write it on the paper.

Nayaka: Okay. What do you think the overall purpose of this prototype is, and who do you think would use it?

Kai: I think it is like a digital classroom. It could be for students or teachers. Maybe it would be useful if something similar to COVID happened again and people had to learn from home.

Nayaka: Yes. One of the ideas behind this is that during COVID, students could sometimes find it difficult to stay focused when classes were only through something like Zoom. With this, students could feel more like they are actually inside a classroom.

Kai: Yeah.

Nayaka: From one to five, how difficult do you think the prototype is to use?

Kai: It is not very difficult because I am using a mouse. If I was using a VR controller, I do not think it would be much harder either. It is pretty easy overall.

Moderator note: The session itself was completed using the VR headset and physical controllers. The wording above is retained from the original recording transcript.

Nayaka: Thinking specifically about the VR controllers you are using now, did they feel easy to control?

Kai: Yeah. They are easy to use and both controllers respond properly.

Nayaka: Would you use something like this if you were in a situation where you needed to attend classes remotely?

Kai: Yeah. The idea is cool, especially because you are inside a classroom instead of being alone at home. I think I would definitely use it.

Nayaka: Thank you. Do you have any feedback or anything you think I should improve?

Kai: It seems pretty good. I think everything is pretty clear. I am just trying to think of something. The writing and deleting controls are clear. I do not really have anything else to add. It is pretty good.

Nayaka: Okay. Thank you so much for your time.

Kai: No worries.

P02 — Syifa: camera-recorded transcript

Nayaka: Okay, I am going to do VR testing. May I know your name?

Syifa: Syifa.

Nayaka: Can I record this interaction?

Syifa: Yeah.

Nayaka: Okay. First, can you grab the two controllers? What do you see on the screen?

Syifa: There is a pop-up guide.

Nayaka: Before we continue, can you see both physical controllers inside the VR environment?

Syifa: Yeah.

Nayaka: Are both controllers following your hands correctly?

Syifa: Yeah.

Nayaka: What does it say?

Syifa: Controls.

Nayaka: Okay. Press A. The A button is on your right controller. What can you understand from the guide?

Syifa: There are instructions for the controls.

Nayaka: So you understand the guide?

Syifa: Yeah.

Nayaka: Can you move from one table to another table?

Syifa: Like this?

Nayaka: Yes. Maybe close the guide first and then try moving to another table.

Syifa: Okay.

Nayaka: Did both physical controllers stay aligned with your hands when you moved?

Syifa: Yeah.

Nayaka: Can you look around? What do you see on the table?

Syifa: Paper.

Nayaka: What else?

Syifa: The instructions.

Nayaka: Can you also see the teaching board from where you are sitting?

Syifa: Yeah.

Nayaka: Can you look between the teaching board and the paper comfortably?

Syifa: Yeah.

Nayaka: Can you copy the date, horizontal line, and circle from the board onto your paper?

Syifa: Okay.

Nayaka: Was that comfortable?

Syifa: Yeah.

Nayaka: Can you try writing something on the paper? You can look at the guide again if you need help.

Syifa: Okay.

Nayaka: Now try changing the tool to the eraser.

Syifa: Okay.

Nayaka: Can you erase what you wrote?

Syifa: Yeah.

Nayaka: What other button can you see?

Syifa: Clear.

Nayaka: Can you try clearing the paper?

Syifa: Okay.

Nayaka: Now can you move to the whiteboard?

Syifa: Okay.

Nayaka: Nice. I can see you are looking at the guide again. You can close it when you are ready. What do you see on the whiteboard?

Syifa: The tools and colours.

Nayaka: You can move around using both controllers. Try moving forward. Now can you write something on the whiteboard using the red marker? You can check the guide again if you are confused.

Syifa: Okay.

Nayaka: Nice. What else can you see on the whiteboard interface?

Syifa: The colours.

Nayaka: What else?

Syifa: The tools. Eraser.

Nayaka: Anything else?

Syifa: Record.

Nayaka: Can you try the Record button? Before that, can you clear the whiteboard?

Syifa: Okay.

Nayaka: Now press Record and write something on the whiteboard. Maybe use the blue marker. Select Marker first, then select Blue.

Syifa: Okay.

Nayaka: Nice. Now press Stop and then try the Play button.

Syifa: Okay.

Nayaka: Let us try it again. Draw something, but make sure you press Record first. After you finish, press Stop, and then press Play.

Syifa: Okay.

Nayaka: What do you understand from this activity?

Syifa: We can write something on the whiteboard, draw something on the paper, and record it as well.

Nayaka: Did the replay match what you expected?

Syifa: Yeah.

Nayaka: And while moving and drawing, did the left and right VR controllers stay responsive?

Syifa: Yeah, both of them worked properly.

Nayaka: Yes. The purpose is to create a virtual classroom so that students who are studying remotely, such as through Zoom, can feel more like they are inside a real classroom and hopefully pay more attention to the teacher. Would you use something like this if you were studying remotely, like during COVID?

Syifa: Yeah.

Nayaka: How would you rate the interaction activities?

Syifa: Ten.

Nayaka: Ten out of ten?

Syifa: Yeah.

Nayaka: Do you have any feedback or anything you think should be improved?

Syifa: No, it is already good.

Nayaka: Okay. Thank you for your time.

Appendix D — Success-measure audit and evidence limits

Planned measure	Target	Available evidence	Audit status
Shared seated view	$\geq 4/5$	5 task passes; five ratings of 5/5	Met
Paper tool discovery	$\geq 4/5$ within 15 s, no direct hint	5 task passes; exact time/prompt level absent	Functional result met; timing unverified
Paper drawing	$\geq 90\%$ acceptable required strokes	5 task passes; five reliability ratings of 5/5; no defect count	Supported; exact percentage unverified
Lesson-to-notes	$\geq 4/5$ and comfort 4–5	5 passes; five comfort ratings of 5/5	Met
Navigation stability	Tracking stable; $\geq 4/5$ rating 4–5	5 passes; five ratings of 5/5	Met
Whiteboard recording	$\geq 4/5$ complete and explain replay	5 passes; recorded evidence for P01/P02	Met
Simulator controller choice	T/Y/Escape every attempt	All sessions were headset sessions	Not applicable / not evaluated
Overall confidence	Median $\geq 4/5$; no severe issue	Median 5/5; zero recorded severe issues	Met

Methodological limitations

- Only two sessions have audio/video recordings; three rely on observation notes.
- The facilitator sometimes gave task prompts or demonstrations, which may increase completion and ratings.
- All six rating measures reached the maximum, creating a ceiling effect that reduces sensitivity to smaller usability differences.
- The supplied records do not include exact discoverability timing, per-stroke defect counts, or systematic assistance coding.
- The convenience sample and brief session support a usability decision for this prototype, not a claim that the concept improves learning outcomes.

Traceability note

The complete combined transcript/results record is stored at Design Evaluation/Interview Transcripts/Iteration2_XRStudyWhiteboard_Transcript_and_Results.md. The original testing plan remains recoverable through Git history. The assessment brief and rubric supplied by the user were used as evaluation criteria and are not treated as participant evidence.